ZEGUAN WU

CONTACT INFORMATION

University of Pittsburgh
210 S. Bouquet Street
Pittsburgh, PA 15260

Email: zew79@pitt.edu

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RESEARCH INTERESTS

Quantum Computing, Optimization, AI, and their intersections

EDUCATION

Lehigh University — Bethlehem, PA
Ph.D., Industrial and Systems Engineering — 2025
Advisor: Tamás Terlaky & Xiu Yang
Committee: Arielle Carr, Carleton Coffrin, and Michael Saunders

Columbia University — New York, NY
M.Sc., Operations Research — 2019

Nanjing University — Nanjing, China
B.Sc., Material Physics — 2018

WORK EXPERIENCE

University of Pittsburgh — Pittsburgh, PA
NASA Postdoc Fellow — 2025 - Present

Pacific Northwest National Laboratory — Richland, WA
Quantum Computing Internship — 2025 - 2025

Los Alamos National Laboratory — Los Alamos, NM
Graduate Research Assistant — 2023 - 2025

PAPERS UNDER REVIEW

1. "Provably efficient quantum algorithms for solving nonlinear differential equations using multiple bosonic modes coupled with qubits", Yu Gan, Hirad Alipanah, Jinglei Cheng, **Zeguan Wu**, Guangyi Li, Juan José Mendoza-Arenas, Peyman Givi, Mujeeb R Malik, Brian J McDermott, Junyu Liu.
2. "*Stab-QRAM: an all-Clifford quantum random access memory for special data*", Guangyi Li, Yu Gan, **Zeguan Wu**, Xueyue Zhang, Zheshen Zhang, and Junyu Liu.
3. "*An efficient quantum algorithm for linear system problem in tensor format*", **Zeguan Wu**, Sidhant Misra, Tamás Terlaky, Xiu Yang, and Marc Vuffray.

PUBLICATION

1. "*Preconditioned quantum interior point method for linear optimization*", Computational Optimization and Applications, **Zeguan Wu**, Xiu Yang, and Tamás Terlaky.
2. "*A quantum dual logarithmic barrier method for linear optimization*", INFORMS Journal on Optimization, **Zeguan Wu**, Pouya Sampourmahani, Mohammadhossein Mohammadisiahroudi, and Tamás Terlaky.

3. "*An inexact feasible interior point method for linear optimization with high adaptability to quantum computers*", SIAM Journal on Optimization, Mohammadhossein Mohammadisiahroudi, Ramin Fakhimi, **Zeguan Wu**, and Tamás Terlaky.
4. "*Optimal Scaling Quantum Interior Point Method for Linear Optimization"*, IEEE Quantum Week 2025, Mohammadhossein Mohammadisiahroudi, **Zeguan Wu**, Pouya Sampourmahani, Jun-Kai You, and Tamás Terlaky.
5. "*Improvements to quantum interior point method for linear optimization"*, ACM Transactions on Quantum Computing, Mohammadhossein Mohammadisiahroudi, **Zeguan Wu**, Brandon Augustino, Arielle Carr, and Tamás Terlaky.
6. "*An inexact feasible quantum interior point method for linearly constrained quadratic optimization*", Entropy, **Zeguan Wu**, Mohammadhossein Mohammadisiahroudi, Brandon Augustino, Xiu Yang, and Tamás Terlaky.
7. "*Quantum-enhanced regression analysis using state-of-the-art QLSAs and QIPMs*", ACM/IEEE Quantum Workshop, Mohammadhossein Mohammadisiahroudi, **Zeguan Wu**, Brandon Augustino, Arielle Carr and Tamás Terlaky.
8. "*Network design for silent link user equilibrium*", Research Reports on Computer Science, Xu Chen, **Zeguan Wu**, and Xuan Di.

CONFERENCE

1. "*Quantum interior points methods for linear optimization*", SIAM-NNP Annual 2024, Rochester, NY.
2. "*A quantum dual logarithmic barrier method for linear optimization*", MOPTA 2024, Bethlehem, PA.
3. "*A quantum dual logarithmic barrier method for linear optimization*", SIAM Annual 2024, Spokane, WA.
4. "*A detailed implementation of QLSA for linear system problem in tensor format*", INFORMS Optimization Society Conference 2024, Houston, TX.
5. "*An inexact feasible quantum interior point method for linear and quadratic optimization*", INFORMS Annual 2023, Phoenix, AZ.
6. "*An inexact feasible quantum interior point method for linear and quadratic optimization*", MOPTA 2023, Bethlehem, PA.
7. "*An inexact feasible quantum interior point method for linearly constrained quadratic optimization*", IISE Annual 2023, New Orleans, LA.
8. "*An inexact feasible quantum interior point method for linearly constrained quadratic optimization*", Mid-Atlantic NA-Day 2022, Philadelphia, PA.
9. "*A preconditioned quantum interior point method for linear optimization*", Flash Talk, INFORMS Annual 2022, Indianapolis, IN.
10. "*A preconditioned quantum interior point method for linear optimization*", Poster, ICCOPT & MOPTA 2022, Bethlehem, PA.

TEACHING EXPERIENCE

Lehigh University:

2023S ISE 111 Engineering Probability, TA

2022F ISE 406 Introduction to Mathematical Optimization, TA

2022S ISE 305/404 Simulation, TA

2021F ISE 365/465 Applied Data Mining, TA

Others:

2023 Gene Golub SIAM Summer School, Optimization Laboratory Tutorial

ACADEMIC SERVICE

Session Chair, MOPTA, 2024

Session Chair, SIAM Annual, 2024

Session Chair, INFORMS Optimization Society Conference, 2024

Session Chair, SIAM-NNP Annual, 2023, 2024

Session Chair, INFORMS Annual, 2023, 2024

Staff, Gene Golub SIAM Summer School, 2023

Session Chair, IISE Annual, 2023

Vice President, Lehigh University INFORMS Student Chapter, 2022-2023

Volunteer, ICCOPT Conference, 2022

Reviewer for European Journal of Operational Research

Reviewer for IEEE Transactions on Quantum Engineering

Reviewer for Journal of Computational Science

Reviewer for Optimization and Engineering

Reviewer for Scientific Reports

Reviewer for IEEE Quantum Week workshop

AWARD

Los Alamos National Laboratory Quantum Computing Summer School Fellowship (2023)

Rossin Professional Development Program (2023)

REFERENCES

Tamás Terlaky	Professor of Industrial and Systems Engineering, Lehigh University
Xiu Yang	Associate Professor of Industrial and Systems Engineering, Lehigh University
Arielle Carr	Assistant Professor of Mathematics, Lafayette College
Marc Vuffray	Staff Research Scientist, Los Alamos National Laboratory
Junyu Liu	Assistant Professor of Computer Science, University of Pittsburgh